

Fast Geometry and Staged Transport for Large LXe TPC Simulation

LXeMC Design Note

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Abstract

This note documents the current design of the LXeMC detector and transport model, using the LZ detector as a concrete example of a large liquid-xenon time projection chamber (LXe TPC). The design goal is not maximal generality. The goal is to simulate very large event samples quickly while retaining the physics that matters for rejection and classification. The resulting approach is a staged simulation: a specialized fast kernel performs most of the transport outside the fiducial volume (FV), where most events are rejected quickly, while a full photon/electron stack is retained only inside the FV, where detailed single-site versus multi-site behavior matters.

1 Introduction

Large LXe TPCs are designed to reject backgrounds aggressively. In a detector such as LZ, most external gammas are rejected by active veto regions before they reach the fiducial volume. This observation drives the central design choice of the present simulation: the transport outside the FV should be optimized for speed and early rejection, while the transport inside the FV should preserve the detailed cascade structure needed for event topology studies.

The code therefore does not attempt to use one uniform transport model everywhere. Instead, it exploits the detector semantics directly. Outside the FV, the simulation only needs enough geometry and physics to decide whether a gamma is vetoed, absorbed in passive material, or survives into the FV. Inside the FV, the simulation keeps the full stack of photons and charged particles because the event classification depends on the spatial pattern of energy depositions.

2 Problem Statement

The practical task is to simulate a very large number of gamma events in a large LXe detector. In the intended use case, the number of events is so large that a uniform full simulation everywhere in the detector is too expensive. This is not primarily a software-engineering inconvenience; it is a physics-rate problem. The simulation must be fast enough to process event counts large enough to support background studies at the level of interest.

The immediate consequence is that the dominant cost must be spent only on the small subset of trajectories that can still affect the final physics observable. Any cost spent on fully tracking obviously rejected events is wasted.

3 Constraints and Boundary Conditions

The design is shaped by the following constraints.

1. **Speed is dominant.** The simulation must run at the level of millions of events per second on representative benchmark configurations.

2. **Most events are rejected early.** For many source configurations, gammas are vetoed in active xenon outside the FV, absorbed in passive material, or degraded to energies irrelevant for the region of interest.
3. **Detailed transport is only needed locally.** The expensive part of the physics—full photon/electron cascades and cluster structure—matters primarily inside the FV.
4. **The detector partition is physically meaningful.** The geometry is not arbitrary. It should reflect the detector regions that carry distinct transport or classification meaning.

4 Design Philosophy

The design is based on a simple principle: use detector semantics to reduce the cost of transport.

The detector is partitioned into regions that matter for either material transport or event classification. The transport kernel is then compiled *ad hoc* from that partition. This is intentionally specialized. The detector geometry is not interpreted as a fully generic tree at runtime in the hot path. Instead, the canonical geometry is loaded once and transformed into a compact fast-kernel representation with explicit scalar region parameters.

This leads to three consequences.

1. The geometry used by the fast kernel is chosen for speed rather than for maximal abstraction.
2. The detector partition is explicit and canonical. It is not inferred dynamically from overlapping semantic flags at runtime.
3. The full transport stack is retained only where the physics demands it, namely inside the FV.

5 Simulation Strategy

The current simulation strategy has three stages.

5.1 Fast rejection

The first goal is to reject events as soon as they become irrelevant. If a gamma deposits visible energy above the veto threshold in the active xenon outside the FV, the event is rejected immediately. If a gamma degrades below the configured ROI floor in passive material, it is terminated locally because it can no longer produce an FV event in the energy region of interest.

5.2 Fast transport outside the FV

Outside the FV, a specialized fast kernel handles gamma transport using a small set of canonical detector regions. This kernel performs:

- point classification into canonical detector regions,
- analytic distance-to-boundary calculations,
- material-aware interaction sampling,
- immediate veto and ROI-floor decisions.

This stage is optimized for the common case in which the gamma is rejected or absorbed before it can matter to the final physics analysis.

5.3 Full transport inside the FV

If a gamma survives into the FV, the fast kernel hands off its state to a full photon/electron transport stack restricted to the FV geometry. This stack retains:

- photon propagation,
- lepton propagation,
- secondaries through the particle stack,
- deposit clustering relevant to event topology.

Inside the FV, the geometry is simple: a homogeneous LXe cylinder bounded by the fiducial radial and longitudinal cuts. The expensive part is therefore not the geometric navigation but the cascade physics itself.

6 Current Status and Benchmark

The current code implements the staged architecture described above.

- The canonical detector geometry is defined explicitly in `detector_lz_v3.json`.
- The fast kernel uses the canonical detector partition outside the FV.
- The FV transport uses a dedicated `FVGeometry` plus a full particle stack.
- The legacy flat-detector workflow has been removed from the canonical code path.

Representative benchmarks place the simulation in the target regime of roughly 10^6 events per second for the fast calibration workflow, with exact values depending on source position and direction. Configurations aimed directly toward the FV are naturally slower than configurations rejected promptly in outer veto regions, but the code remains in the intended high-throughput regime.

7 Geometry Model

7.1 Shared geometric primitives

The file `geometry_core.jl` contains the shared shape and wrapper layer. It provides:

- primitive solids: `Cyl`, `CylShell`, `Box`, `Disk`, `Cap`, `DomedContainer`, `CappedCylinder`;
- logical wrappers: `LCyl`, `LCylShell`, `LBox`, `LDisk`;
- materialized wrappers: `PhysicalVolume`, `PCyl`, `PCylShell`, `PBox`, `PDisk`;
- common geometric operations such as containment tests, masses, and ray entry/exit distances.

These are general geometry tools. They are not tied to the fast kernel or to the FV stack specifically.

7.2 Canonical detector geometry

The file `geometry.jl` now contains the canonical detector geometry representation. It loads the tracking detector from JSON into a tree of `TrackingDetector` nodes, with explicit region semantics and geometry.

The canonical LZ tracking geometry is partitioned into the following regions:

- `AirDome`
- `LXe_passive`
- `TopActive`
- `BarrelActive`
- `BottomActive`
- `FV`
- `Skin`
- `FC_PTFE`
- `FC_rings`

This partition is exhaustive for the intended detector model and reflects the physics logic directly.

7.3 Meaning of the canonical regions

AirDome Gas region above the liquid xenon.

LXe_passive Passive liquid xenon outside the active veto regions and below the cathode.

TopActive Active xenon cylinder between the gate and the top of the FV.

BarrelActive Active xenon shell around the FV over the FV longitudinal span.

BottomActive Active xenon cylinder between the bottom of the FV and the cathode.

FV Fiducial xenon volume used for event selection and topology.

Skin Active xenon skin outside the field-cage region.

FC_PTFE PTFE part of the field cage.

FC_rings Titanium ring part of the field cage.

7.4 Source geometry

The source geometry is described separately from the detector tracking geometry. This separation is intentional. Source geometry is concerned with radioactive emission locations and source-side transport; detector tracking geometry is concerned with transport and classification once a gamma enters the detector environment of interest.

8 Transport Model

8.1 Fast kernel outside the FV

The fast kernel operates on a compiled geometry representation generated from the canonical detector tree. It performs:

1. region classification,
2. analytic distance-to-boundary calculation,
3. material lookup,
4. interaction-distance sampling,
5. immediate region-dependent decisions.

This kernel treats transport in the outer detector as a problem of fast geometric and material decisions, not as a full generic cascade simulation.

8.2 Full stack inside the FV

The FV transport uses a dedicated `FVGeometry` object compiled directly from the canonical detector geometry. The FV stack keeps:

- photons,
- electrons and positrons,
- bremsstrahlung,
- the particle stack and deposit list.

The geometry inside the FV is reduced to what is physically needed: the LXe material and the cylindrical FV boundary.

8.3 Handoff logic

The fast kernel tracks a gamma until one of the following occurs:

- it is vetoed outside the FV,
- it is absorbed or terminated below the ROI floor in passive material,
- it escapes,
- it reaches the FV and remains relevant.

Only in the last case is the gamma handed to the FV transport stack.

8.4 ROI floor

The simulation uses an ROI floor in gamma energy. Once a gamma energy drops below this threshold, the code assumes that it can no longer generate a relevant contribution to the final ROI and terminates it locally. This is a deliberate analysis-motivated approximation used to gain speed.

8.5 Veto logic

The fast kernel uses region-dependent visible-energy thresholds:

- active xenon outside the FV: veto at the configured TPC threshold;
- skin: veto at the configured skin threshold;
- passive regions: no veto, only transport and possible local termination.

9 Event Logic

9.1 Calibration sampling

The calibration path samples a small multiplicity of gammas and injects them with controlled energy, position, and direction. This path is used for benchmarks, controlled studies, and debugging.

9.2 Veto conditions

Deposits in active xenon outside the FV veto the event when they exceed the configured threshold. This allows the event to be rejected immediately without continuing detailed transport.

9.3 FV acceptance and multi-site rejection

Once in the FV, the event is classified using the deposit pattern returned by the full stack. The key observable is whether the deposit pattern is consistent with a single-site or multi-site event under the configured clustering model.

9.4 Role of `dz_resolution`

The FV clustering logic uses a longitudinal clustering scale `dz_resolution`. Deposits separated beyond this scale contribute to multi-site structure, while deposits consistent within this scale may be merged into a single-site candidate.

10 Code Map

The current code ownership is as follows.

geometry_core.jl	shared geometric primitives, wrappers, and geometric helpers.
geometry.jl	canonical detector geometry, tracking tree, region semantics, fast-kernel compilation, FV geometry compilation.
tracking.jl	fast transport outside the FV, FV-only full stack, event-level logic.
source.jl	source-side propagation inside source volumes.
decays.jl	decay and calibration sampling.
scripts/test_event_processing.jl	canonical benchmark and event-processing driver.
scripts/geometry_viewer.html	canonical detector and source geometry viewer.

The public entry points of primary interest are:

- `load_tracking_detector`
- `compile_fastkernel_geometry`
- `compile_fv_geometry`
- `process_event`
- `process_event_fastkernel_calib`
- `propagate_gamma_in_fv`

11 Current Limitations

The current simulation is fast and purposeful, but it is not intended to be an arbitrary general detector framework. The main limitations are:

1. The fast kernel is specialized to the canonical detector partition.
2. The ROI-floor termination is an intentional approximation.
3. The outside-FV transport is optimized for rapid rejection rather than for full generic cascade tracking in every material.
4. The benchmarked fast path is best suited to the current detector semantics and event logic, not to arbitrary future geometries without recompilation of the fast-kernel model.

These limitations are acceptable because they follow directly from the project goal: simulate extremely large event samples as fast as possible while retaining the detailed physics only where it matters.